

Anthony Paredes-Bautista

773-610-4769 | anthonyparedesb0@gmail.com | [linkedin.com/in/anthonypb](https://www.linkedin.com/in/anthonypb) | github.com/Anthony-PB | [portfolio](#)

EDUCATION

Cornell University - College Of Engineering

Ithaca, NY

Bachelor of Science in Computer Science, Minor in Artificial Intelligence

*Expected Graduation Date: **May 2027***

- Intro CS: Design & Development, Object-Oriented Programming and Data Structures, Intro to Android Development, Foundations of AI Reasoning and Decision-Making, Introduction to Analysis of Algorithms, Embedded Systems, Intro to Machine Learning, Data Structures and Functional Programming, Intro to Game Architecture, Intro to Deep Learning

TECHNICAL SKILLS

Languages: Ocaml, Python, Java, JavaScript, TypeScript, HTML/CSS, Assembly, Kotlin
Frameworks & Libraries: React.js, Django, Express.js, JUnit, NumPy, Pandas, PyTorch, SQLite, FireBase
Tools & Platforms: Node.js, MongoDB, Docker, Git

PROJECTS

Automato: Video Knowledge Aggregator *FastAPI, React Native, TypeScript, Supabase*

November 2025

- Built React Native frontend for an AI-powered app that transcribes short-form videos and returns structured notes, implementing a video modal, animated sidebar, and category filter system.
- Extended FastAPI backend to return Claude-generated content categories and video URLs; integrated API responses with frontend data layer. (**Claude Builders Club Hackathon 2025**)

Human Protein Atlas Classification *Numpy, Python, Tensorflow, Pandas*

January 2025

- Built a machine learning pipeline to classify protein microscopy images using data from the Human Protein Atlas.
- Preprocessed multi-channel images with custom one-hot encoding and trained a convolutional neural network using TensorFlow, optimizing data pipelines with `tf.data`.
- Evaluated model performance via validation accuracy, achieving **95.72%**.

RoomFind *React, TypeScript, Node.js, Express, Firebase Firestore*

April 2025

- Built a Cornell-gated roommate matching platform with a 4-factor weighted compatibility algorithm (noise, cleanliness, sleep schedule, pets) that scores each listing 0–100 against a user's saved preferences and sorts results accordingly.
- Engineered client-side Canvas image compression (800×600, 72% JPEG) to eliminate server-side processing, integrated Nominatim geocoding with debounced autocomplete, and used Firestore batch writes for atomic cascading deletes on account removal.

3D N-Body Gravity Simulation *OCaml, Raylib, Dune*

December 2025

- Built real-time 3D physics sandbox with gravitational interactions, collision detection, body merging, and trails.
- Engineered custom physics engine from scratch using only OCaml standard library, implementing Euler integration and Newton's gravitational law.

Prakriti: Feeding the Future *Arduino, React, JavaScript, HTML, CSS, Netlify*

February 2024

- Developed software modules and deployed the website for Prakriti, a prototype IoT solution to curb rice crop burning in India (responsible for 86 million tonnes of CO₂ emissions annually).
- Collaborated on design and development, helping secure a **\$3,000** award at the Cornell Digital Ag Hackathon for our team of 6.

CAMPUS INVOLVEMENTS

- Cornell Cybersecurity Club (Main Club): Active member preparing for competitions. On the SWE track.
- Fitness Enthusiast(Cornell Barbell Club): Regularly work out to maintain health, and I intend to participate in a powerlifting competition.

COURSES

- CYB101 — Intro to Cybersecurity (CodePath Spring 2025)
- CYB102 — Intermediate Cybersecurity (CodePath Summer 2025)
- TIP102 — Intermediate Technical Interview Prep (CodePath Fall 2025)